



Practice Problem Set -04 (Indeterminate Forms)

1. $\lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{x^2 \sin x}$ Ans: $-1/3$

2. $\lim_{x \rightarrow 0} \frac{x \cos x - \log(1+x)}{x^2}$ Ans: $1/18$

3. $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$ Ans: 2

4. $\lim_{x \rightarrow 0} \frac{x e^x - \log(1+x)}{\cosh x - \cos x}$ Ans: $3/2$

5. $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x} \right)$ Ans: $-1/3$

6. $\lim_{x \rightarrow 0} (a^x + x)^{\frac{1}{x}}$ Ans: ae

7. $\lim_{x \rightarrow 0} (1 + \sin x)^{\cot x}$ Ans: e

8. $\lim_{x \rightarrow 0} \frac{\cot x - 1/x}{x}$ Ans : $-1/3$